# docs_sect_8_4_mars_MOC_NA

Processed on: 2025-10-31 11:54:20

ASP version: 3.6.0-alpha

asp_plot version: 1.9.0

## DEM Summary

| Property | Value |
|---|---|
| DEM File | output-DEM.tif |
| Dimensions (px) | 1236 x 6576 |
| GSD (m) | 6.95 |
| CRS | EPSG:None |
| Nodata (%) | 70.0 |
| Elevation Range (m) | 3648.9 to 6104.1 |

# Input Scenes

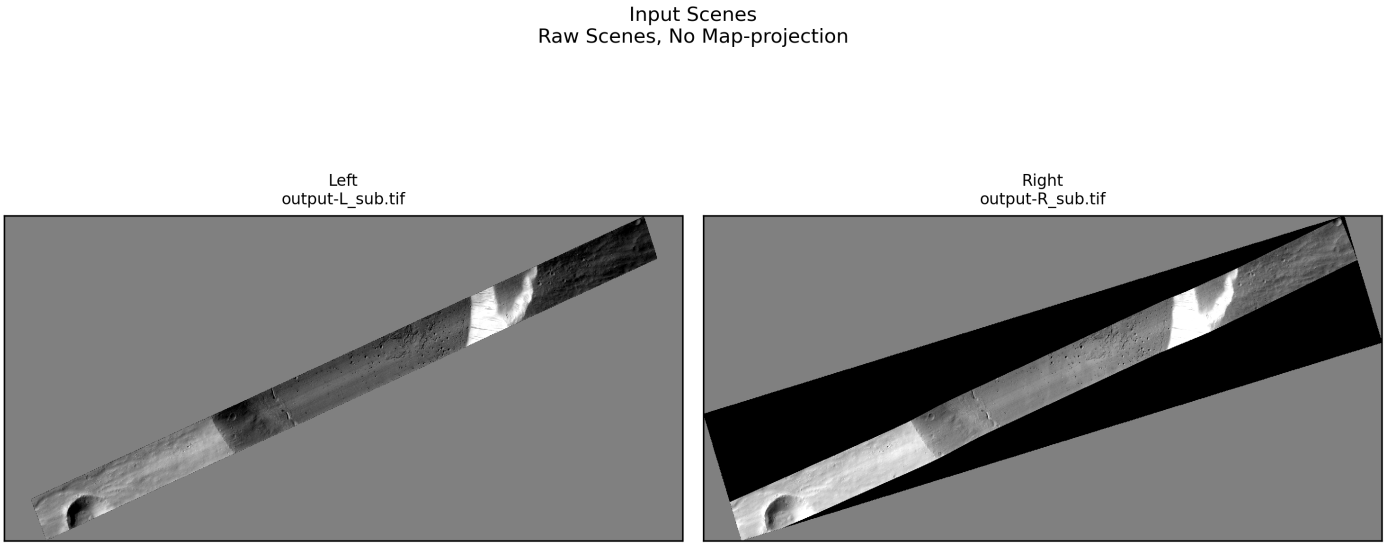


*Figure 1: Left and right input scenes used for stereo processing. Non-mapprojected scenes are shown after ASP's alignment step (e.g., affineepipolar), which rotates images to create horizontal epipolar lines for correlation. Mapprojected scenes require no pre-alignment and are displayed in their map-projected orientation.*

# Match Points

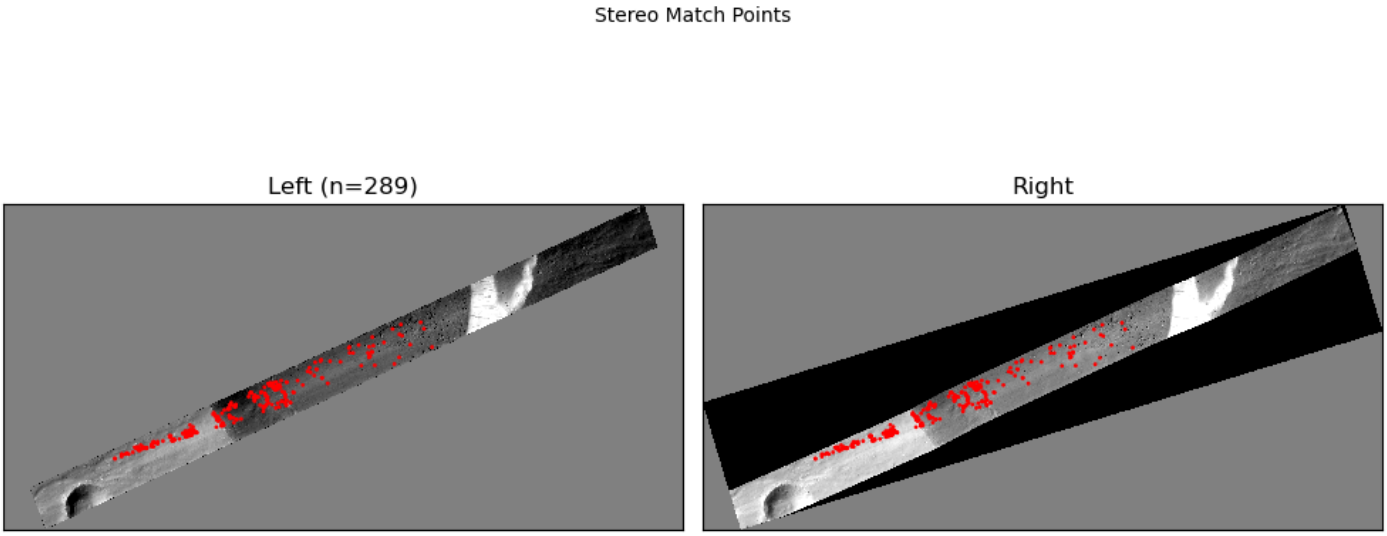


*Figure 2: Interest point matches between left and right images identified during stereo correlation.*

# Detailed Hillshade

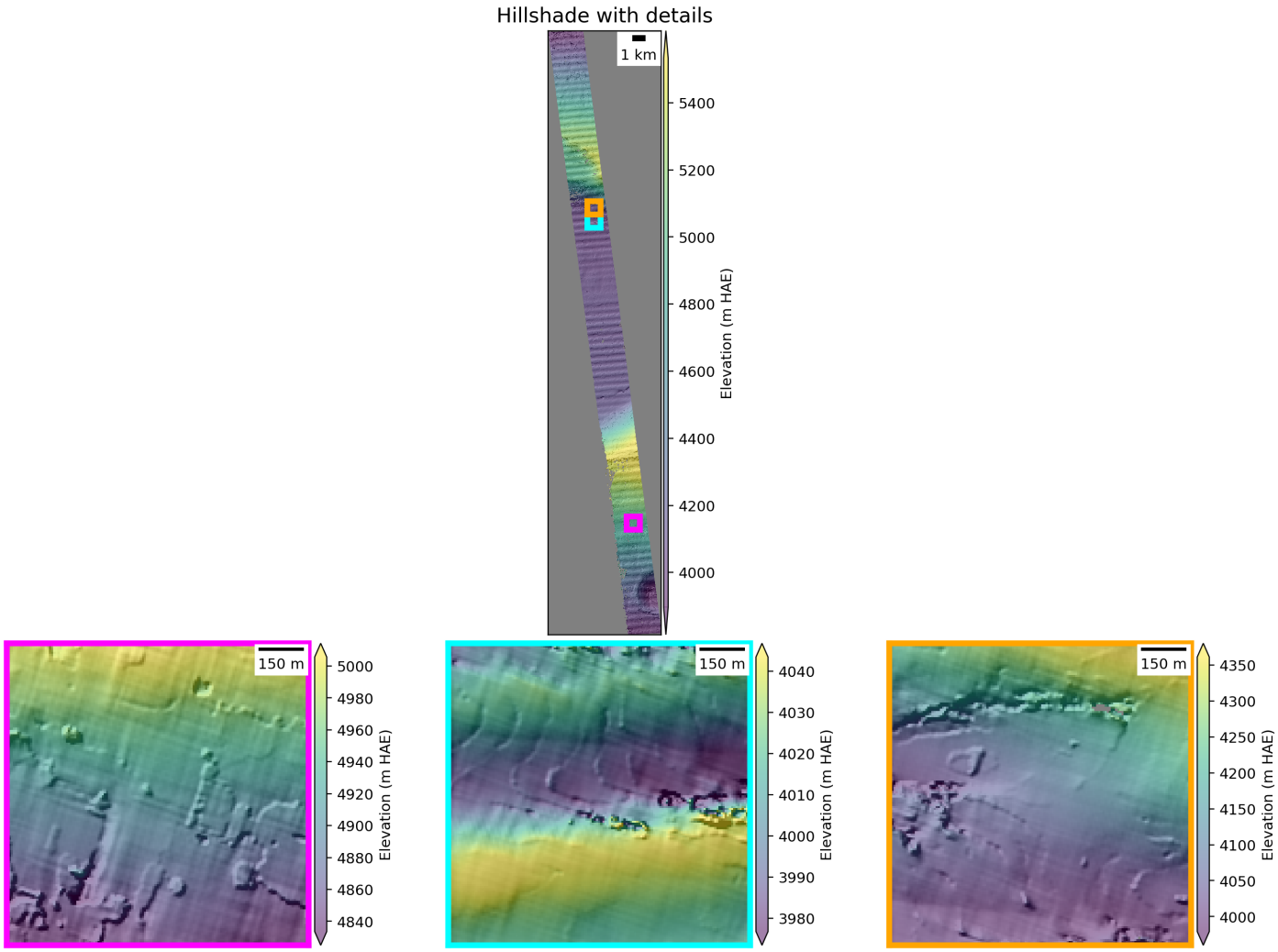


*Figure 3: DEM hillshade. If the intersection error is available, zoomed subsets selected from low, medium, and high (left to right) uncertainty areas are displayed in the second row. If the mapprojected image is available, corresponding ortho image subsets are displayed in the bottom row.*

# DEM Results

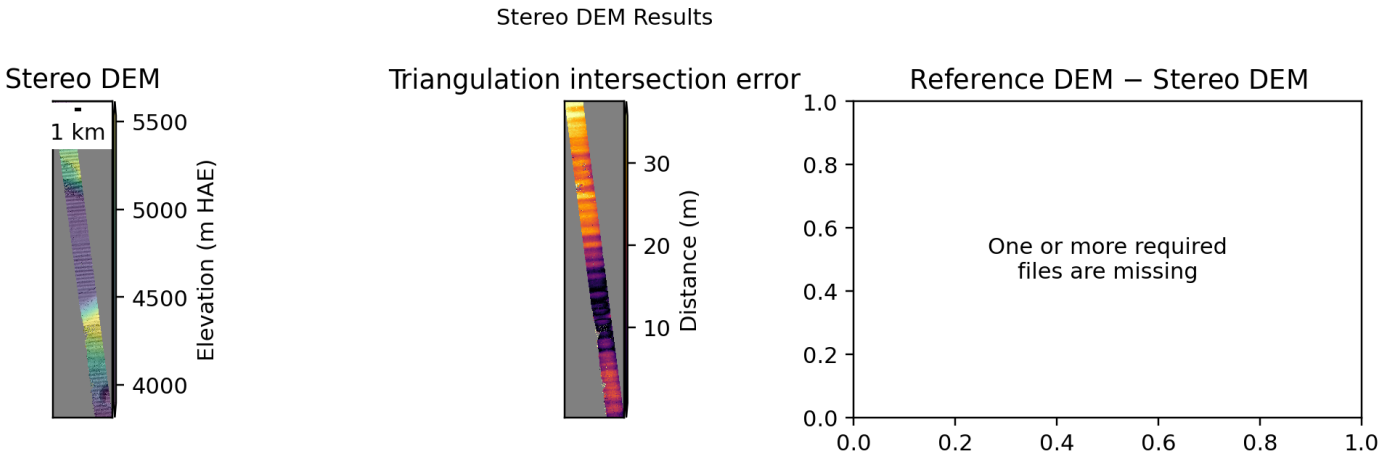


*Figure 4: Output DEM with intersection error map and difference relative to the reference DEM used in processing.*

# Disparity

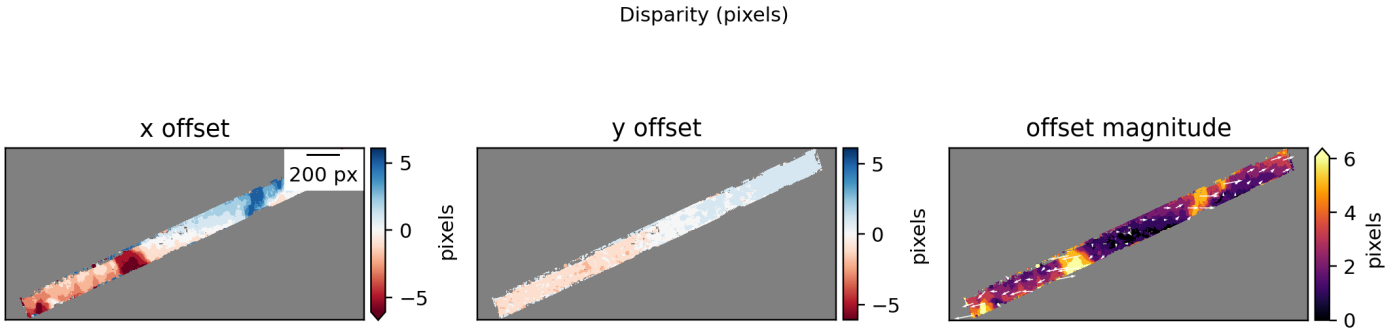


*Figure 5: Horizontal and vertical disparity maps in pixels with quiver overlay.*

## MOLA Altimetry Map

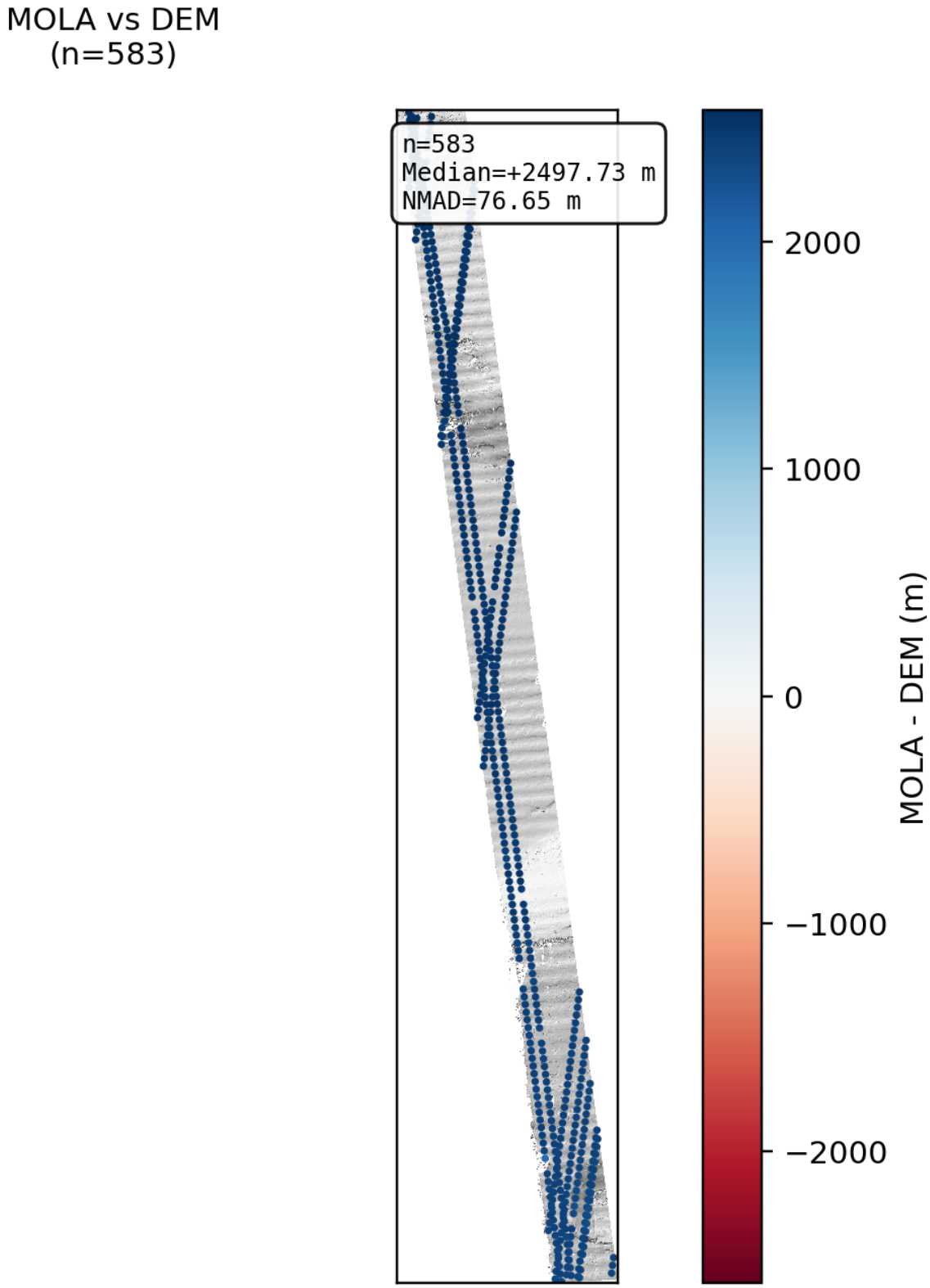


*Figure 6: MOLA elevation differences vs. ASP DEM.*

## MOLA Altimetry Histogram

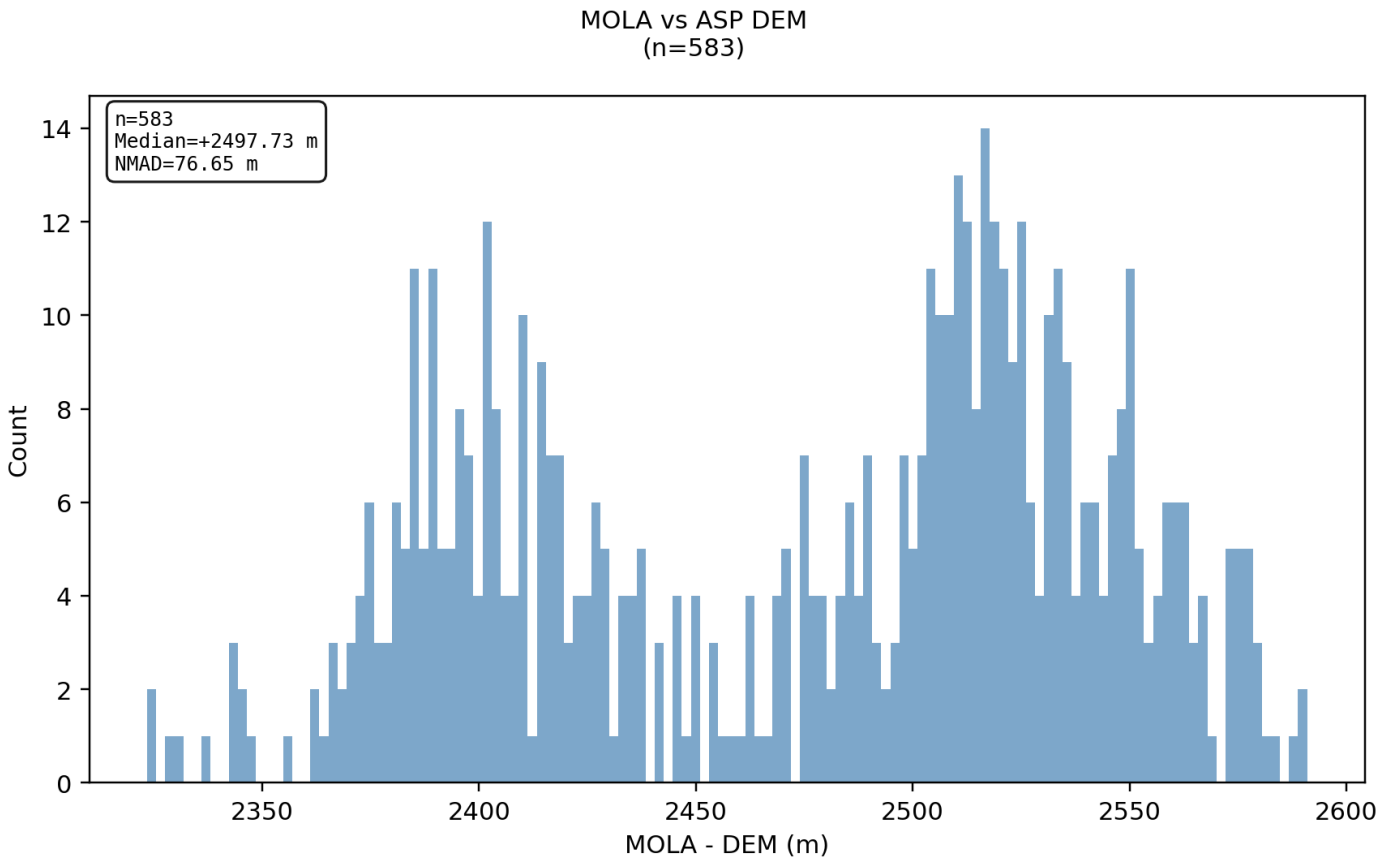


*Figure 7: Distribution of elevation differences between MOLA and ASP DEM.*

# Processing Parameters

## Runtime Summary

| Step | Runtime |
|---|---|
| Bundle Adjust | N/A |
| Stereo | 0 hours and 2 minutes |
| point2dem | 0 hours and 0 minutes |

### Bundle Adjust Command:

```
Bundle adjustment not run
```

### Stereo Command:

```
stereo M0806047.map.cub R0701361.map.cub result/output --corr-seed-mode 1 --sgm-collar-size 0 --compute-point-cloud-center-only --threads 8
```

### point2dem Command:

```
point2dem -r mars --stereographic --auto-proj-center result/output-PC.tif --errorimage result/output-PC.tif
```

### Report Generation Command:

```
asp_plot --directory /Users/ben/Dropbox/UW_Shean/asp_plot-test_data/docs_examples/docs_sect_8_4_mars_MOC_NA/ --stereo_directory result/ --add_basemap False --altimetry_csv /Users/ben/Dropbox/UW_Shean/asp_plot-test_data/docs_examples/docs_sect_8_4_mars_MOC_NA//438125_mola/MolaPEDR_24N24N_263E263E_20260319T205257836_topo_csv.csv --plot_geometry False
```